Amirreza Amini

Data Scientist | ML & AI Engineering

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PROFESSIONAL SUMMARY

Data scientist with a prior career in instrumentation and control engineering — a background that brings first-hand understanding of how physical processes generate, and corrupt, the data that machine learning systems depend on. MSc Data Science graduate (Best Project Award) with end-to-end experience building, containerising, and deploying ML and NLP systems to production. Proficient in Python, PyTorch, and classical ML, with hands-on work in RAG pipelines, vector databases, and LLM fine-tuning.

TECHNICAL SKILLS

Languages: Python, SQL, R

Machine Learning: Scikit-learn, XGBoost, Imbalanced-learn (SMOTENC), Regression & Classification, Cross-Validation & Hyperparameter Tuning, SHAP (explainability)

Deep Learning & NLP: PyTorch, TensorFlow, HuggingFace Transformers, LLM Fine-Tuning, Neural Networks

Generative AI: RAG Pipelines, Vector Databases (Weaviate), Cohere Reranking, Prompt Engineering

Deployment & MLOps: FastAPI, Streamlit, Docker, AWS (EC2, ECR), Hugging Face Spaces, Git/GitHub

Big Data & Visualisation: PySpark, Tableau, Matplotlib, Seaborn, Altair

Observability: OpenTelemetry, Arize Phoenix

PROFESSIONAL EXPERIENCE

Instrumentation & Control Engineer — Pars Smart Explorers *Feb 2023 – May 2024*

- Configured and installed flow-meter and sensor systems across multiple industrial sites, producing daily structured reports on sensor readings and process deviations.
- Diagnosed process faults through root-cause analysis, signal tracing, and controlled testing, reducing resolution time for recurring failures.
- Built deep practical understanding of how physical processes generate — and corrupt — measurement data, directly informing data-quality and feature-engineering work in ML pipelines.

Swimming Teacher — Better (GLL), North Finchley, London *Dec 2025 – Present*

- Plan and deliver structured, progressive swimming lessons for children and adults, tailored to each learner's ability level.
- Apply safeguarding procedures and site safety standards while maintaining attentive poolside supervision.
- Support day-to-day centre operations, including customer service and accurate record-keeping.

KEY PROJECTS

FinSentinel — Financial Sentiment Classifier *Python · PyTorch · FastAPI · Docker · AWS*

- Fine-tuned ProsusAI/FinBERT on the Financial PhraseBank dataset, achieving weighted F1 = 0.974 — a +10.8 pp improvement over a TF-IDF + Logistic Regression baseline.
- Built a custom WeightedTrainer subclass injecting class-weighted CrossEntropy loss to address 62% class imbalance without oversampling.
- Containerised the model and FastAPI inference API, pushed to Amazon ECR, and deployed to AWS EC2 with a live public endpoint returning per-class confidence scores; 15/15 tests passing with pytest.

Loan Default Risk Predictor *Python · XGBoost · FastAPI · Streamlit*

- Built an end-to-end default-prediction system on 307,000 real loan applications (Home Credit), achieving 0.75 AUC-ROC versus a 0.60 baseline.
- Engineered domain-informed features (credit-to-income ratio, employment anomaly flag), validated as significant predictors via SHAP analysis.
- Tuned the classification threshold from 0.50 to 0.18, lifting default recall from 4% to 40%; deployed via FastAPI and an interactive Streamlit UI on Hugging Face Spaces.

Multi-Strategy RAG Pipeline for Question Answering — *MSc Dissertation* *Weaviate · Llama 3.2 · Cohere*

- Designed and built an end-to-end RAG pipeline combining Semantic (Dense), BM25 (Lexical), and Hybrid retrieval with Cohere reranking to reduce LLM hallucination.
- Instrumented the pipeline with OpenTelemetry and Arize Phoenix to profile latency and automatically evaluate the RAG Triad — Faithfulness, Relevance, and Answer Accuracy.

EDUCATION

MSc Data Science — Middlesex University, UK *2025 – 2026*

Awarded **Certificate of Best Project** (Top 5 individual projects, Jan 2026). Key modules: Modelling, Regression & Machine Learning; Visual Data Analysis; Applied Data Analytics (Big Data & Cloud).

BSc Electrical Engineering — Shiraz University, Iran *2018 – 2023*

CERTIFICATIONS

Retrieval-Augmented Generation (RAG) — DeepLearning.AI (via Coursera)

Associate Data Scientist — DataCamp